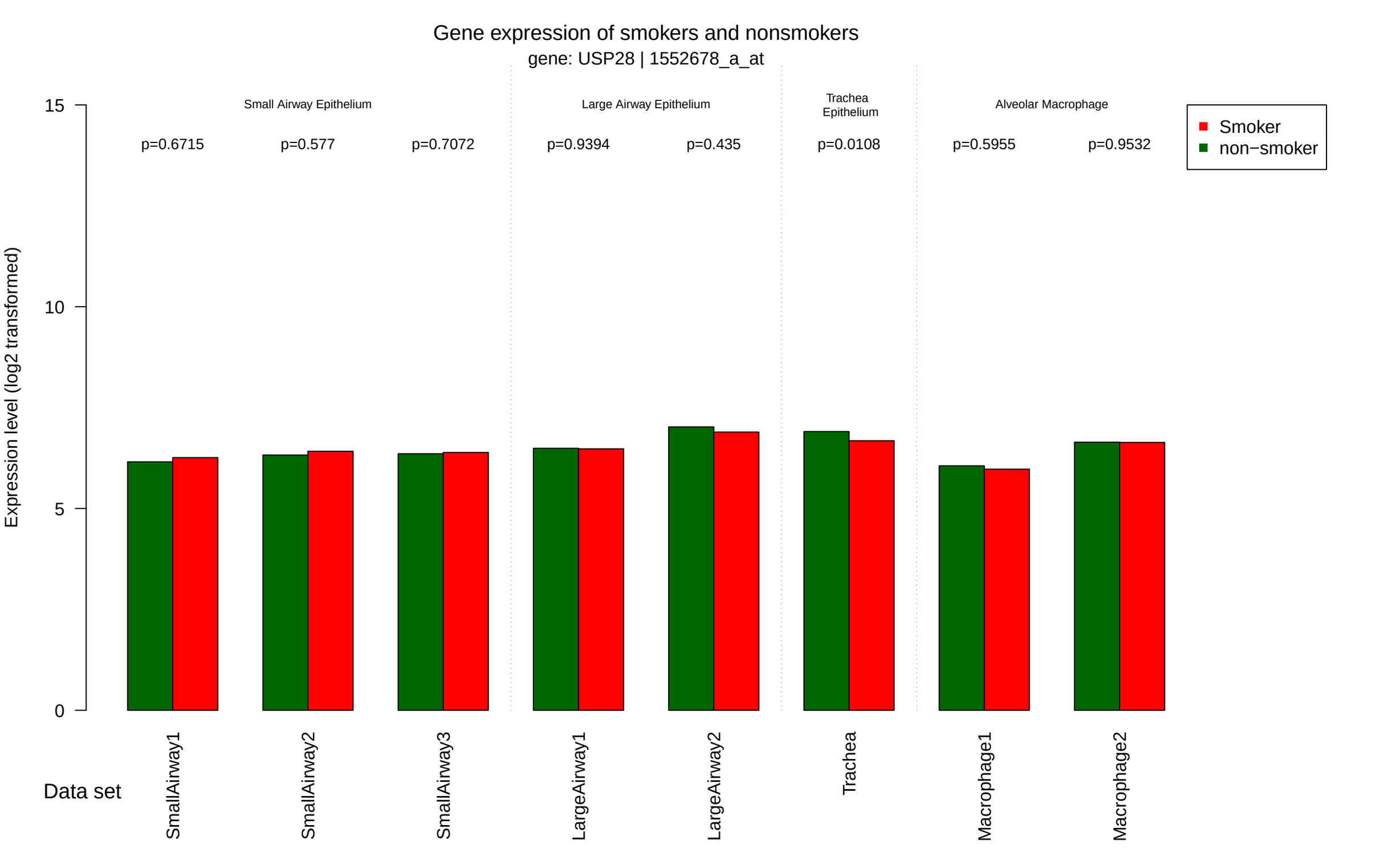
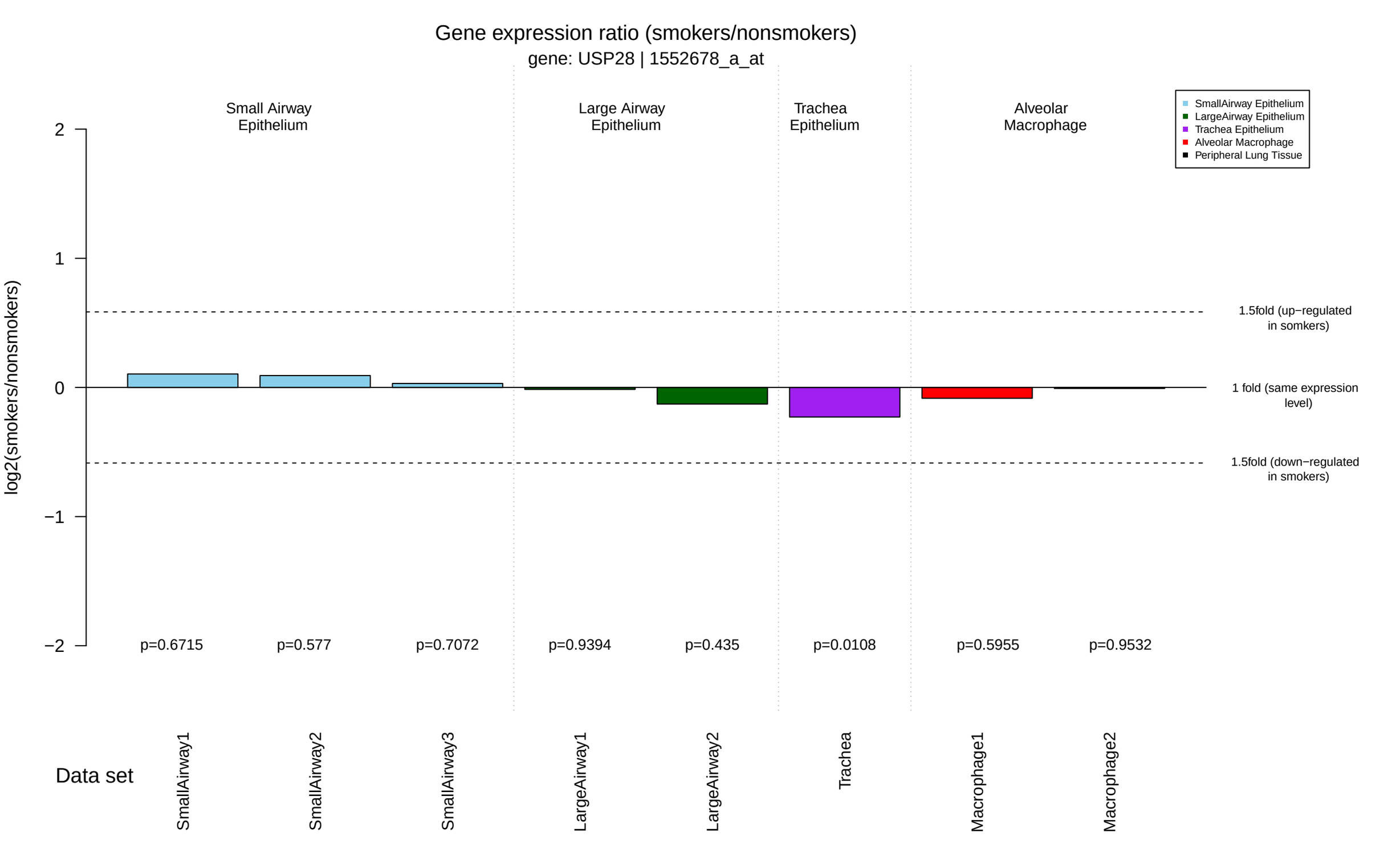
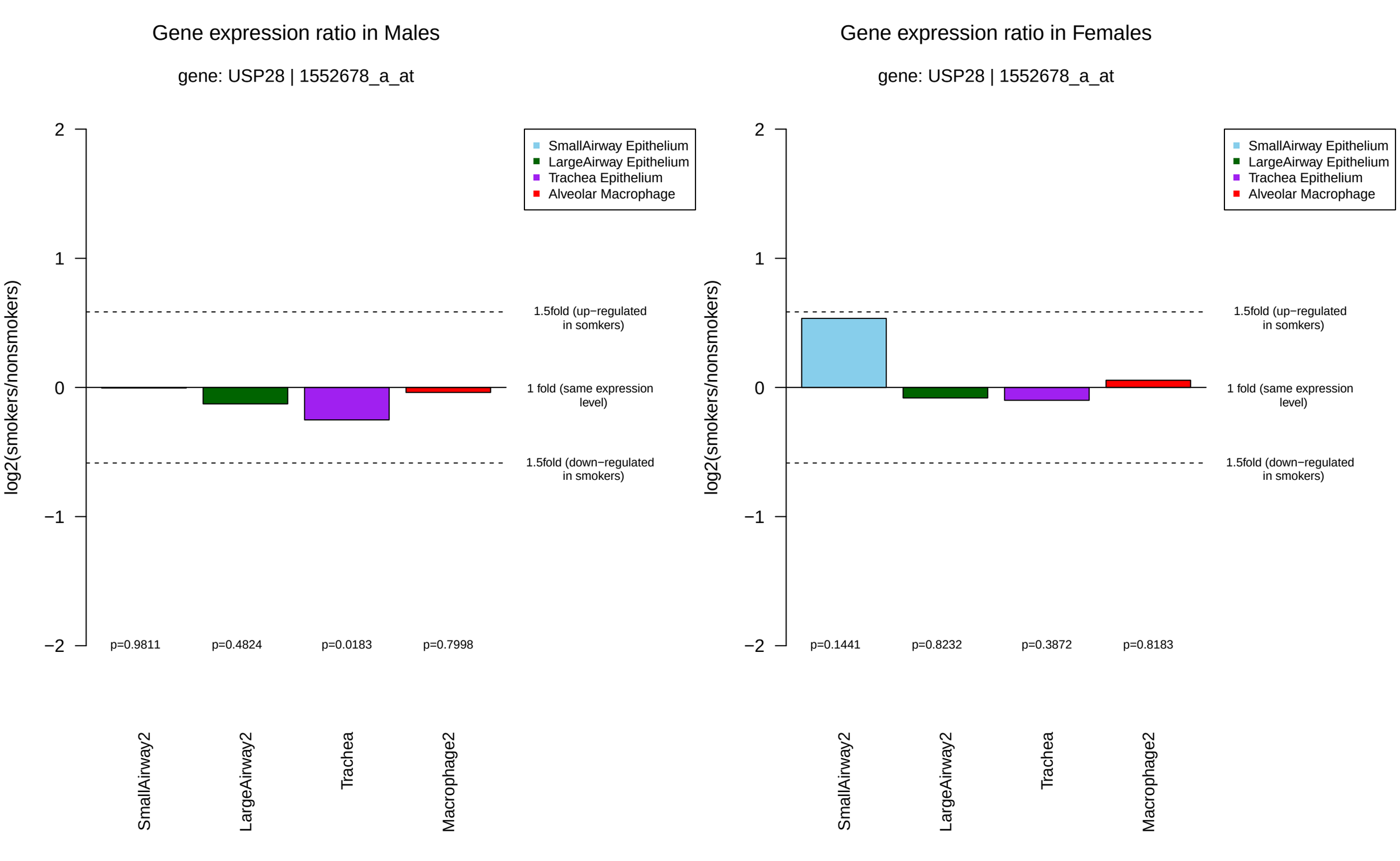
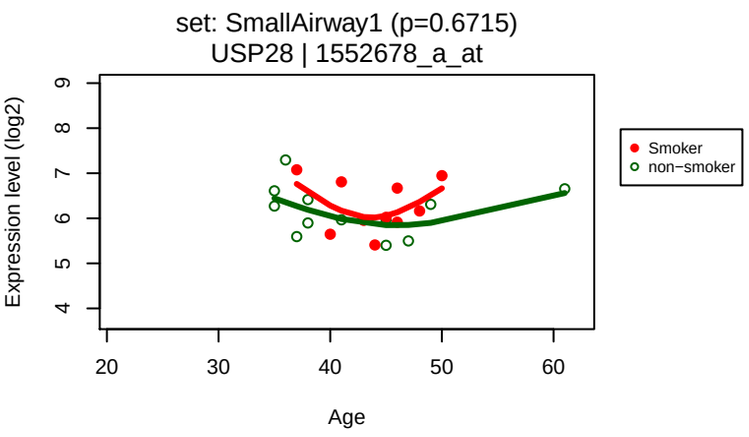
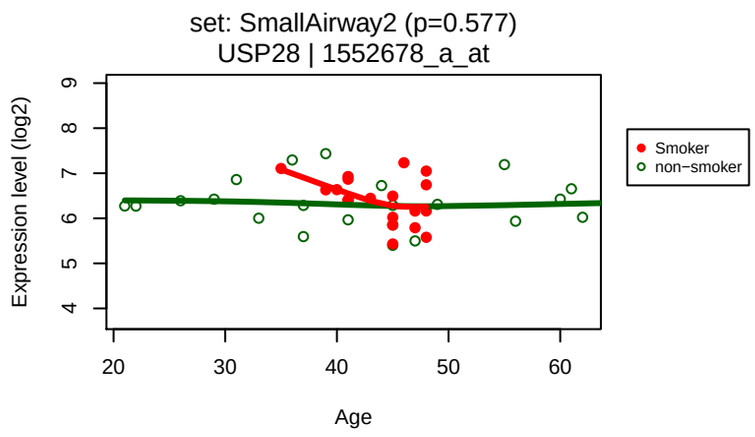
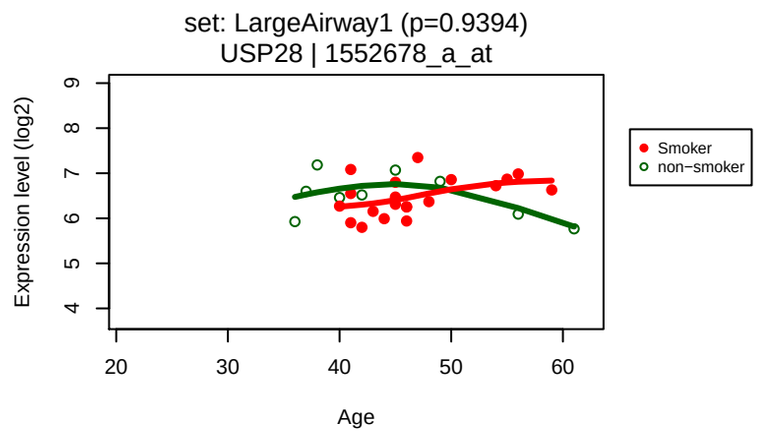

Gene expression of smokers and nonsmokers
gene: USP28 | 1552678_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6715
p=0.577
p=0.7072
p=0.9394
p=0.435
p=0.0108
p=0.5955
p=0.9532
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

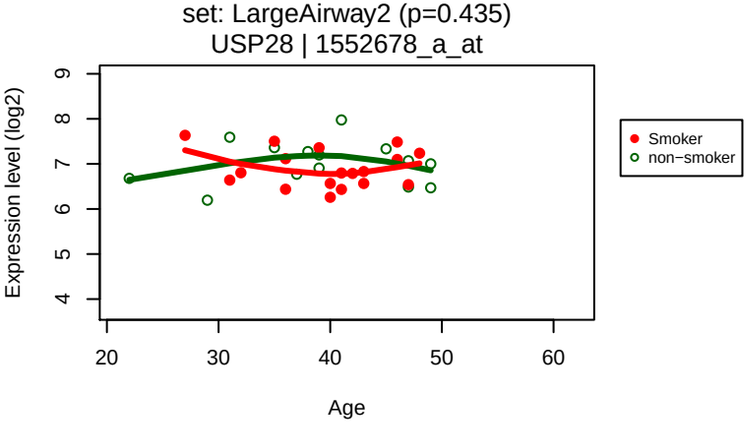

Gene expression ratio (smokers/nonsmokers)
gene: USP28 | 1552678_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6715
p=0.577
p=0.7072
p=0.9394
p=0.435
p=0.0108
p=0.5955
p=0.9532
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

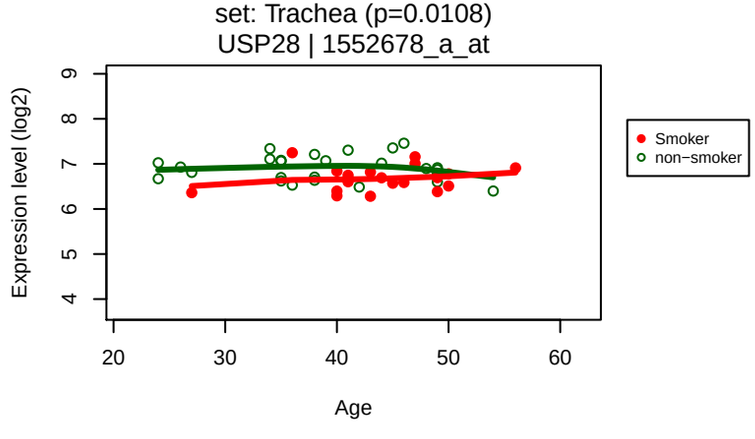

Gene expression ratio in Males
gene: USP28 | 1552678_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9811
p=0.4824
p=0.0183
p=0.7998
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP28 | 1552678_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1441
p=0.8232
p=0.3872
p=0.8183
SmallAirway2
LargeAirway2
Trachea
Macrophage2

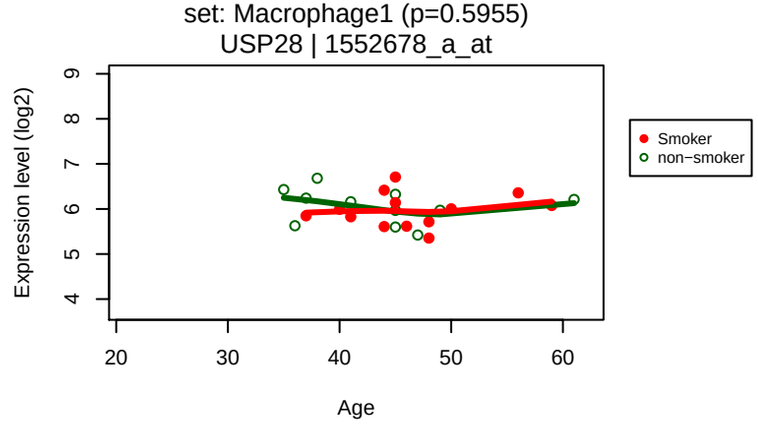

set: SmallAirway1 (p=0.6715)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker

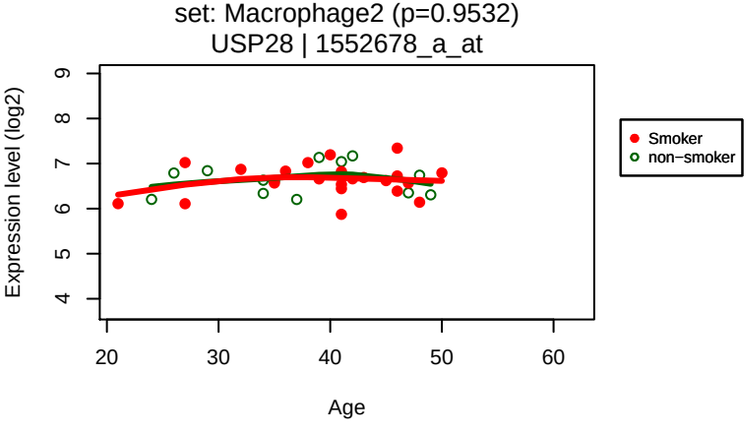

set: SmallAirway2 (p=0.577)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker

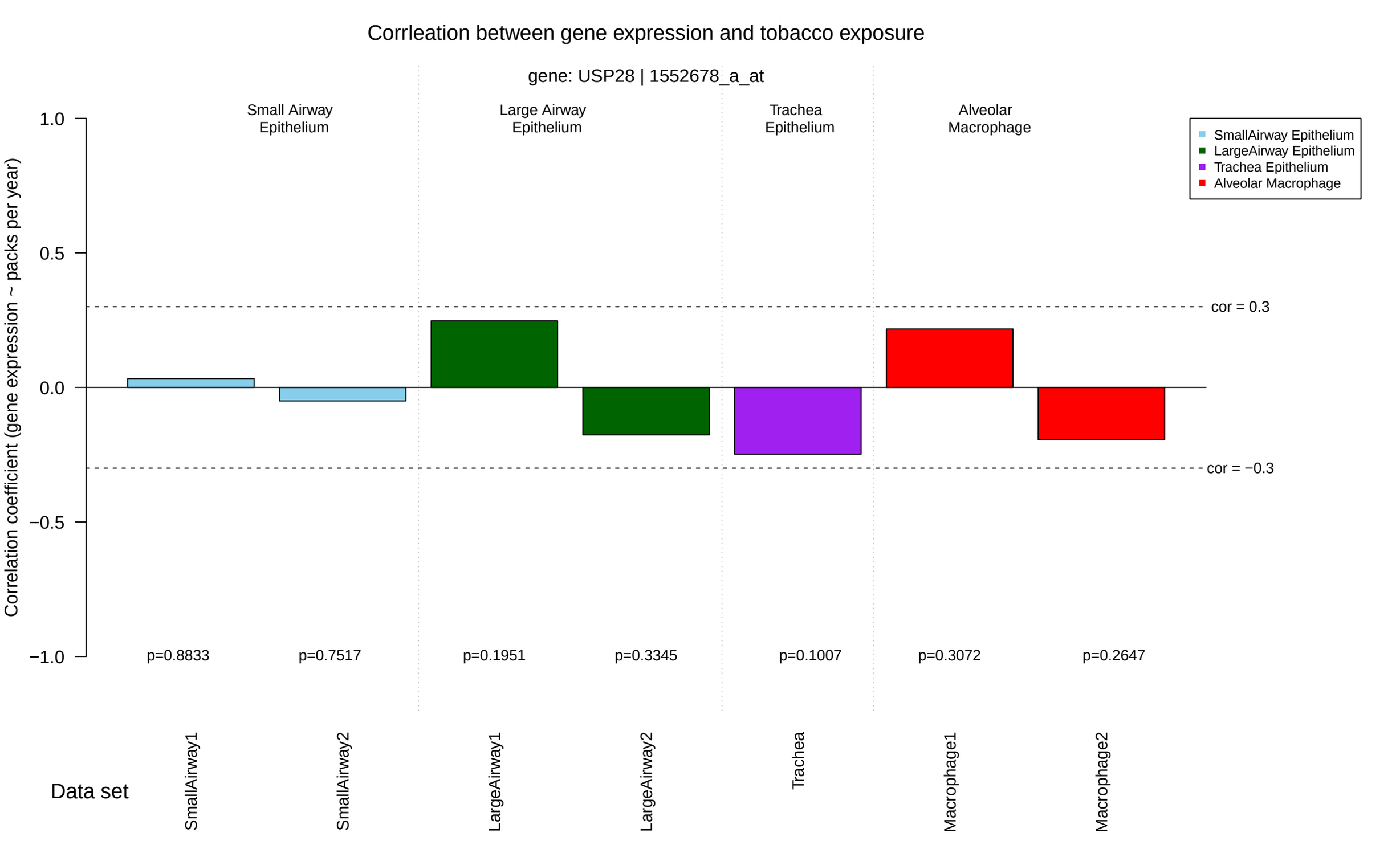

set: LargeAirway1 (p=0.9394)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.435)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0108)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5955)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9532)
USP28 | 1552678_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP28 | 1552678_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8833
p=0.7517
p=0.1951
p=0.3345
p=0.1007
p=0.3072
p=0.2647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2